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| Examination   | University | Institute                     | N Year | CPI/%   |
|---------------|------------|-------------------------------|--------|---------|
| Graduation    | IIT Bombay | IIT Bombay                    | 2028   | 9.59/10 |
| Intermediate  | TSBIE      | Sri Chaitanya Junior Kalasala | 2024   | 97.2%   |
| Matriculation | BSEAP      | Sri Viswa Shanthi High School | 2021   | 100%    |

## SCHOLASTIC ACHIEVEMENTS

- Member of the **North Mumbai Zonal Toppers team** at **Mimamsa**, organised by IISER Pune (2026)
- Secured **5th rank** at IITB and **35th rank** globally in the **Simon Marais Mathematics Competition** (2025)
- Honored with the **Institute Academic Award** for outstanding performance among **1400+ students** (2025)
- Awarded prestigious **Reliance Foundation Merit Scholarship** for outstanding undergraduate studies (2024)
- Secured **All India Rank 10** in the **IIT Joint Entrance Examination Advanced** among 2 lakh candidates (2024)
- Achieved **State-wide Rank 12** in **AP EAMCET Exam 2024** out of **280,000+ candidates** state-wide (2024)

## OLYMPIADS

- Among the top 35 students in India to clear the **Indian National Physics Olympiad (INPhO)** and invited **twice** to attend the **Orientation-Cum-Selection Camp for International Physics Olympiad (IPhO)** (2023,24)
- Awarded **Gold Medal** at **Orientation-Cum-Selection-Camp (OCSC)** for International Physics Olympiad (2023)
- Ranked among **Top 29** participants nationwide in **INAO** and was invited to **Astronomy OCSC** for IOAA (2024)
- Secured **All India Rank 4** in the **International Earth Science Olympiad (IESO)** in 8-th grade (2020)
- Awarded the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY)** Fellowship in 2021–2022 (2022)

## QUANT EXPERIENCE

### Quantitative Research Intern | Quadeye

(May 2026 – July 2026)

- Exploring **ensemble pruning** and model compression for large-scale tree ensembles to **reduce inference time** without losing accuracy and f1, while building **reproducible pipelines** for pruning, validation and comparison
- Evaluated pruned ensemble models and neural networks for **correctness, runtime, scalability and generalizability**

### Optiver Future Focus | Mumbai

(Jun 2026)

- Studied **market making, options pricing, volatility, and algorithmic-trading** through lectures and workshops
- Applied concepts in **structured trading games, commodity mocks, and Optibook simulations** to analyze risk, execution, pricing inefficiencies, and strategy design under competitive conditions and **real market constraints**

### Jane Street SEE | Hong Kong

(Dec 2025)

- Studied **probability, market structure, and arbitrage** through lectures emphasizing quantitative reasoning
- Applied concepts in **mock trading games** based on real life market data to analyze risk, execution, pricing inefficiencies, and basic arbitrage opportunities under competitive conditions and **realistic market constraints**

## RESEARCH INTERNSHIPS

### Using Lean to Teach Proof Writing (SUAMI) | Carnegie Mellon University

(May - July 2025)

Guide: Prof. Pavel Kovalev

- Developed an **interactive Lean 4 game** using the Lean4 Game Engine, modeled after the early chapters of An Infinite Descent into Pure Mathematics-Clive Newstead, allowing players to improve their **formal reasoning skills**
- Project serves as an optional study supplement and **under consideration by CMU** for integration into courses
- Implemented lightweight **custom tactics** in Lean emulating textbook strategies such as contradiction, contrapositive, introduction and elimination rules allowing beginners to **focus on ideas** rather than Lean syntax in the game

### Exploring the Kinematics and Substructure of M31 & M32 (SIP) | UCSC

(May - July 2024)

Guide: Prof. Raja GuhaThakurta, Rohit Raj

- Used data analysis libraries (NumPy/Pandas/Astropy) to process **SPLASH (Keck/DEIMOS)** and **DESI** survey data of **M31 (Andromeda)** and its satellite **M32**, designing methods to reduce noise caused by stellar clustering
- Evaluated **calibration accuracy** by comparing DESI data with the **HST-catalog** - the most precise in that region
- Built a comparison pipeline to conduct comparison of SPLASH and DESI measurements of velocity and intensity, aimed at understanding **M31 and M32** in the broader context of **investigating the kinematics and substructures**

## PUBLICATIONS

### Solving the N-Queens Problem Using Simulated Annealing | [link]

(2024)

- Published research presenting a novel hybrid algorithm combining **simulated annealing** with **genetic operators** to solve the classical **N-Queens problem**, providing a unique optimization perspective in algorithmic development
- Demonstrated improved performance over classical backtracking by applying **probabilistic and evolutionary methods** to minimize conflicts efficiently while solving N-Queens, ensuring scalability and efficiency of the solutions
- Applied initialization, **iterative annealing** and **fitness-based selection** for scalable higher order N-Queens

## KEY PROJECTS

### Dynamic Routing Engine | Course Project: Data Structures and Algorithms Lab (November 2025)

Guide: Prof. Ashutosh Gupta

- Engineered a **graph routing system** using **K-D tree**, **A\***, and **Yen's K-shortest path algorithms** to handle real-time navigation, traffic updates, and dynamic constraints across **100k+ node** with sub-second response times
- Developed intelligent delivery scheduling solutions for **NP-hard routing problems** using constraint-based heuristics, achieving **30% faster** delivery completion through optimal order batching and fleet management optimization
- Built C++ architecture with **JSON-based APIs** supporting **Euclidean/Manhattan/shortest-path metrics**, traffic-aware routing with **96-interval** speed profiles, and **10-20x faster** approximate queries with **95% accuracy**

### Chess Engine with Classical Game Theory + NNUE | Seasons of Code (May - July 2025)

- Developed a chess engine in C++, performed **Min-max search with Alpha-Beta pruning** to increase search depth
- Trained a **NNUE (Efficiently Updatable Neural Network)** model on extensive board-value data from Stockfish
- Integrated **NNUE with iterative deepening search**, Zobrist hashing and transposition tables to increase efficiency and achieve rating of 2700, demonstrating strong competitive play and consistent performance against skilled players

### Facial Recognition System | Winter in Data Science (December 2024)

- Built a **facial recognition system** with 97% accuracy using TensorFlow for deep learning and OpenCV for efficient real-time image capturing, GPU-accelerated pre-processing, optimized feature extraction, and latency reduction
- Performed **HOG-based detection**, extracted 128D embeddings and applied **distance-weighted k-NN**, combined with pre-processing (grayscale, resizing, normalization), PCA and evaluation on labeled datasets for accuracy

## OTHER PROJECTS

### xv6 Kernel Extensions | Course Project: OS, Guide: Prof. Purushottam (Jan 2026 - Apr 2026)

- Implemented **kernel-level multithreading** in xv6 using clone and join, enabling independently schedulable threads to share an address space and file descriptors while maintaining separate stacks, trapframes, execution state
- Implemented **futex-based synchronization** in xv6, including sleeplocks and semaphores, and applied them to a **producer-consumer** and **reader-writer** workloads using **sleep/wake** mechanisms to eliminate busy waiting
- Enhanced the xv6 scheduler with **priority scheduling**, **CPU affinity**, and **context-switch tracking**

### Distributed Algorithms Simulation Project | IIT Indore, Prof. Debasish (Dec 2025)

- Extended an existing **distributed-algorithm simulator** (based on the CCM Model) by implementing a new algorithm for **coordinated distribution of k agents across a graph**, where agents operate independently
- Implemented multiple strategies for agent dispersion, compared their **efficiency** through large scale experiments
- Built **modular simulation components** enabling rapid prototyping of new agent distribution algorithms

### Circuit Synthesis Using SAT Solver | Course Project: Logic, Guide: Prof. Krishna S (August 2025)

- Built a SAT-solver based framework to automatically **synthesize NAND-only circuits** from arbitrary Boolean formulae/ truth tables while **minimizing the number of gates** through iterative deepening on number of gates
- Implemented in **Python** with PySAT, encoded input formulas into truth tables and applied NAND logic to generate valid NAND connections with constraints for input encoding, gate count, logic semantics and output constraints

### Assorted Projects | Self Projects (2023-Present)

- Built a two-player **Angry Birds-style game** in **pygame-ce** with real physics, wind, themes and custom power-ups
- Built a SAT solver in c++ using DPLL/CDCL with clause learning, intelligent backtracking, and efficient heuristics
- Built a **number recognition system** entirely in **NumPy**, implementing forward/backward propagation and the **Adam optimizer**, without any external ML libraries to train and evaluate a small neural network from scratch
- Developed a **Rubik's Cube solver** in Python using the IDA\* algorithm with a 54-element state, misplacement-based heuristic, move-cycle tables, pruning of inverse/duplicate states and depth-bounded search for efficient solutions
- Implemented a **Sudoku solver** using recursive backtracking with constraint checks and logging for real-time steps
- Designed a **spell-checker** with a Trie-based dictionary, enhanced **Levenshtein distance** using keyboard proximity and frequency-weighted ranking for accurate, context-aware auto-correction and word auto-complete suggestions
- Built a **MCMC** tool for decrypting basic substitution encryption and evaluating performance on noisy text data

## TECHNICAL SKILLS

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| Languages      | Python, Bash, HTML, CSS, JavaScript, Make, Matlab, Assembly, Verilog, MIPS                                  |
| Software/Tools | Git, GitHub, Jupyter, L <sup>A</sup> T <sub>E</sub> X, Sed, Awk, Fusion360, Flask, FastAPI, SQLite, Astropy |
| Data Science   | NumPy, Pandas, Matplotlib, TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn                                 |

## EXTRACURRICULAR ACTIVITIES

- Selected for Phase 2 of **Graviton Crucible**, learning market making through trading games and lectures (2026)
- Participated in **ACM Summer School 2026** at IIT Bombay on Foundations of **Hardware Security** (Jun 2026)
- Designed and built a **quadcopter** from the ground up, including a custom joystick PCB using Arduino, frame modeling with Fusion 360, laser-cut fabrication and integrated a PID controller to achieve stable control (2025)
- Participated in the **NSO athletics** training program, developing discipline, endurance and teamwork skills (2025)
- Awarded full scholarship by **Sri Chaitanya School**, with stipend support to join advanced training (2024)
- Achieved **third place** in **Tyro CTF Competition** across all undergraduates (IIT Bombay) (2024)
- Trained under-resourced students** in Mathematics and Science for 2 years, improving local literacy (2020-2024)
- Taught Yoga and Meditation to local community kids, promoting health, mindfulness and awareness (2020-2024)